👩🏾‍🏫 Staff Guide · Horticulture 1 · Unit 7 Propagation (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
Her readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as her worksheets for that lesson
- Quick check: score, then scale to her test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Textbook Unit 7**: No textbook scan for Unit 7; notes are team-written from her slides (only 2 slides had text).
- **Propagation quiz**: No class quiz in the export; our quiz is team-written.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

✂️ Propagation: Guided Notes

Horticulture 1 · Unit 7 · Level 3 (closer to the class work) · goes with Propagation Methods, Vegetable Propagation worksheet, Propagation Journal

Name: _____

Use the class slides and your reading. Key words:

propagation | *sexual* | *asexual* | *clone* | *node* | *stem cutting* | *leaf cutting* | *callus* | *separation* | *division* | *rooting hormone* | *runner*

A. Types

1. Fill in the table.

	Sexual	Asexual
Uses		
Parents		
Offspring		

B. Methods

2. Cut stem cuttings below a _____.

3. Rooting hormone contains _____.

4. Explain the difference between separation and division.

_____

_____

C. Vegetables

5. Fill in the table.

Vegetable	Propagated by
Potato	
Garlic	
Strawberry	
Sweet potato	

✂️ Propagation: Reading

Horticulture 1 · Unit 7 · Level 3 (closer to the class work) · goes with Propagation Methods, Vegetable Propagation worksheet, Propagation Journal

Name: _____

Making New Plants

Propagation is the making of new plants. Sexual propagation uses seeds, which form when pollen from one parent fertilizes the egg of another. The new plants vary, just as brothers and sisters differ. Asexual propagation uses parts of one plant, such as stems, leaves or roots. The new plant is a clone, an exact genetic copy of the parent.

Growers use asexual propagation when they want every plant to look the same. In a stem cutting, a 3-4 inch piece is cut just below a node, the lower leaves are removed, and the end is dipped in rooting hormone, a powder containing the plant hormone auxin. The cutting is stuck in moist mix and kept humid until roots form in two to four weeks.

Some plants grow from leaf cuttings. Succulent leaves are left to dry, or callus, for a few days and then laid on mix, where roots and a tiny plant form. Separation removes natural offsets, such as the 'chicks' around a hen-and-chicks plant. Division splits a clump, like an iris or hosta, into pieces that each have roots and shoots.

Many vegetables are propagated asexually: potatoes from seed pieces with eyes, garlic from cloves, strawberries from runners and sweet potatoes from slips. Rooting hormone and sharp tools require care: wear gloves, pour a little hormone into a cup rather than dipping into the jar, cut away from the body with an adult present, and wash hands afterward.

Check your reading

1. Compare sexual and asexual propagation.

2. List the steps for taking a stem cutting.

3. Why would a grower clone a plant instead of using seeds?

✂️ Cuttings & Separation: Lab Sheet

Horticulture 1 · Unit 7 · Level 3 · goes with Propagation Methods (hen & chicks, leaf and stem cuttings)

Name: _____

Question: Does rooting hormone help stem cuttings root faster?

Materials

6 coleus/geranium cuttings, rooting hormone, pots with moist mix, labels

⚠️ Safety

- Nitrile gloves; hormone from a cup only
- Pruners with an adult; cut away from body
- Wash hands

Prediction

Will cuttings with hormone root faster? Explain.

_____

_____

Procedure

1. Take 6 cuttings below a node.
2. Dip 3 in hormone; leave 3 untreated.
3. Stick all in moist mix; label.
4. After 2-3 weeks, gently check roots and count/measure.

Data

Treatment	Cuttings	Rooted	Longest root (cm)
Hormone	3		
No hormone	3		

Analysis (Claim, Evidence, Reasoning)

1. Claim, evidence, reasoning: did hormone help?

✂️ Propagation: Quick Check

Horticulture 1 · Unit 7 · Level 3 (closer to the class work) · goes with Propagation Methods, Vegetable Propagation worksheet, Propagation Journal

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. A clone is…

A. A plant from two parents | B. A seed | C. An exact genetic copy

2. Stem cuttings are cut just below a…

A. Node | B. Flower | C. Root

3. Splitting an iris clump is…

A. Separation | B. Layering | C. Division

4. Strawberries spread by…

A. Runners | B. Cloves | C. Eyes

5. Why are lower leaves removed from a cutting?

6. Give two safety steps for using rooting hormone.

✂️ Propagation: Guided Notes · ANSWER KEY

Horticulture 1 · Unit 7 · Level 3 (closer to the class work) · goes with Propagation Methods, Vegetable Propagation worksheet, Propagation Journal

Name: _____

Use the class slides and your reading. Key words:

propagation · *sexual* · *asexual* · *clone* · *node* · *stem cutting* · *leaf cutting* · *callus* · *separation* · *division* · *rooting hormone* · *runner*

A. Types

1. Fill in the table.

	Sexual	Asexual
Uses	**seeds**	**stems, leaves, roots**
Parents	**two**	**one**
Offspring	**vary**	**clones**

B. Methods

2. Cut stem cuttings below a **node** .

3. Rooting hormone contains **auxin** .

4. Explain the difference between separation and division.
Separation removes natural offsets; division cuts a clump into rooted pieces.

C. Vegetables

5. Fill in the table.

Vegetable	Propagated by
Potato	**seed pieces with eyes**
Garlic	**cloves**
Strawberry	**runners**
Sweet potato	**slips**

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

✂️ Cuttings & Separation: Lab Sheet · ANSWER KEY

Horticulture 1 · Unit 7 · Level 3 · goes with Propagation Methods (hen & chicks, leaf and stem cuttings)

Name: _____

Question: Does rooting hormone help stem cuttings root faster?

Materials

6 coleus/geranium cuttings, rooting hormone, pots with moist mix, labels

⚠️ Safety

- Nitrile gloves; hormone from a cup only
- Pruners with an adult; cut away from body
- Wash hands

Prediction

Will cuttings with hormone root faster? Explain.

Yes; auxin speeds root formation.

Procedure

1. Take 6 cuttings below a node.
2. Dip 3 in hormone; leave 3 untreated.
3. Stick all in moist mix; label.
4. After 2-3 weeks, gently check roots and count/measure.

Data

Treatment	Cuttings	Rooted	Longest root (cm)
Hormone	3	**3**	**4**
No hormone	3	**1**	**1**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim, evidence, reasoning: did hormone help?

Hormone cuttings rooted more and longer because auxin promotes root growth.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

✂️ Propagation: Quick Check · ANSWER KEY

Horticulture 1 · Unit 7 · Level 3 (closer to the class work) · goes with Propagation Methods, Vegetable Propagation worksheet, Propagation Journal

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. A clone is…

A. A plant from two parents | B. A seed | **C. An exact genetic copy**

2. Stem cuttings are cut just below a…

A. Node | B. Flower | C. Root

3. Splitting an iris clump is…

A. Separation | B. Layering | **C. Division**

4. Strawberries spread by…

A. Runners | B. Cloves | C. Eyes

5. Why are lower leaves removed from a cutting?

They would rot in the moist mix and lose water.

6. Give two safety steps for using rooting hormone.

Wear gloves, pour into a cup, don't touch face, wash hands.

🔑 *Team-written from her Propagation Methods slides and Vegetable Propagation worksheet, please check.*